

# **SMART SOLAR EV CHARGING WITH DUAL BATTERY SWITCHING AND SOLAR TRACKING**

## **A PROJECT REPORT**

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**BONAFIDE CERTIFICATE**

This is to certify that the project report entitled "**SMART SOLAR EV CHARGING WITH DUAL BATTERY SWITCHING AND SOLAR TRACKING**" submitted by

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in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Communication Engineering is a Bonafide record of the work carried out under my guidance and supervision at Amrita School of Engineering, Bengaluru.

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**EXAMINER 1**

**EXAMINER 2**

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## ABSTRACT

With the rise in global EV adoption, sustainable, intelligent and autonomous charging solutions are required to support EVs without relying on traditional grid power and fossil fuels. This project design and implementation a Smart Solar EV Charging system based on ESP32 microcontroller which incorporates a Dual Battery Switching feature and an Automatic Solar tracking feature. It combines three key functional areas: autonomous solar energy harvesting using a four-LDR sun-tracking system, intelligent dual-battery management with relay based power source switching and remote rover control using a web-enabled interface.

The solar tracking subsystem uses 4 Light Dependent Resistors (LDRs) to be placed at the top, bottom, left and right sections of the solar panel frame. The ESP32 constantly reads data from the LDR and controls two servo motors to dynamically turn the panel to the light source with the highest intensity, thereby capturing the maximum amount of PV energy for the entire day. The battery management subsystem monitors the voltage of the main battery (2S lithium-ion) using a resistive voltage divider, and the ADC of the ESP32. Once the battery voltage drops below a predetermined level (6.5 V), the system automatically switches onto a relay to use the auxiliary battery which is charged by the solar panels, keeping the load running without causing deep discharge of the main battery. Real-time data of the system, such as battery voltage, battery charge percentage (shown as a graphical bar), whether a power source is active or not, and whether the motor is moving or not is displayed on a web-based dashboard hosted on the ESP32. There are also direction control buttons on the dashboard for manually operating the rover. The 16×2 I2C LCD can be used for local display of important parameters to complement the remote monitoring. Results indicate that the solar tracker is operating correctly, the switching between batteries is working correctly and that the web-based control is responding quickly, providing an effective, cost effective and scalable solution for solar assisted EV charging with IoT control.

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## **LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE**

|              |                                                     |
|--------------|-----------------------------------------------------|
| <b>ADC</b>   | Analogue-to-Digital Converter                       |
| <b>BMS</b>   | Battery Management System                           |
| <b>DC</b>    | Direct Current                                      |
| <b>ESP32</b> | Espressif Systems 32-bit Wi-Fi + BT Microcontroller |
| <b>EV</b>    | Electric Vehicle                                    |
| <b>GPIO</b>  | General-Purpose Input/Output                        |
| <b>I2C</b>   | Inter-Integrated Circuit                            |
| <b>IoT</b>   | Internet of Things                                  |
| <b>JSON</b>  | JavaScript Object Notation                          |
| <b>LCD</b>   | Liquid Crystal Display                              |
| <b>LDR</b>   | Light Dependent Resistor                            |
| <b>MPPT</b>  | Maximum Power Point Tracking                        |
| <b>PCB</b>   | Printed Circuit Board                               |
| <b>PWM</b>   | Pulse-Width Modulation                              |
| <b>SoC</b>   | State of Charge                                     |
| <b>V2G</b>   | Vehicle-to-Grid                                     |
| <b>Wi-Fi</b> | Wireless Fidelity                                   |
| <b>WPT</b>   | Wireless Power Transfer                             |

## CHAPTER 1 INTRODUCTION

### 1.1 Overall System / Scenario

The world has made a big shift towards electric vehicles in the last ten years, with concerns about climate change, air pollution, and the depletion of fossil fuels fueling this shift. Electric vehicles (EVs) have become a key component of sustainable transportation policies in numerous Asian, European and North American countries, and global EV sales reached 14 million vehicles in 2023 alone. Two challenges still persist despite the significant advancements in battery life and drivetrain efficiency that make EVs more feasible, Range anxiety (the need to avoid running out of charge before arriving at a charging station or a location) and the reliance on grid powered charging stations which may not always be available in rural areas, semi-urbanized areas or off-grid settings. The challenges require innovative solutions to capture and store energy from renewable resources on the vehicle itself, without the need of fixed charging stations .

PV is an attractive energy harvesting option for on-vehicle applications. The sunlight captured by solar panels on the roof, hood or bonnet of a car can be used to boost the main battery pack, decreasing the need to charge from the electricity grid during the day. The research indicates that a well-oriented 1-2 m<sup>2</sup> solar panel on a passenger car can produce 100-300 Wh per day under average conditions of solar radiation, which represents a significant contribution to the battery's capacity even though it is a small fraction of this. It has been observed that for average insolation the potential of a well-oriented solar panel of 1-2 m<sup>2</sup> on a passenger car can reach 100-300 Wh/day, a value significant for short commuters and fleet vehicles in sunny regions, although it is a small fraction of the battery's capacity. The key to this benefit maximisation is the solar tracking system that continuously orients the panel to face the sun's position, despite the diurnal arc of the sun and the orientation of the vehicle. Fixed-panel installations have an energy loss of 20-40 % when compared to single-axis tracker and 35-55 % when compared to dual-axis

tracker, making active tracking a high valued add even in space-limited vehicle-mounted applications.

Along with the harvesting of energy, intelligent battery management is crucial for the safe and efficient operation of the envisaged dual power-source architecture in this project. For a two-battery system, one the main/traction battery and the other an auxiliary battery powered directly from the solar panel, a smart switching controller is needed which will control the flow of electricity to the vehicle load from the most suitable source at the right moment. If the main battery is not intelligently managed, it can suffer deep discharge (which can result in permanent loss of capacity to lithium-ion batteries) and the energy stored in the auxiliary battery can become wasted or used inefficiently. Instead of a complex and expensive bidirectional DC-DC converter system, using a relay based switching circuit controlled by firmware running on a microcontroller is a cost effective approach to this logic implementation.

Currently, the most common and research-based solutions for solar-powered EV charging focus on either the energy harvesting end or battery management end, with few systems offering an integrated supervisory layer that enables EV charging to be remotely monitored and controlled via a web browser. If the operator has no real-time visibility of the battery state, the power source status, the direction of the solar tracking and the movement of the vehicle, from a smartphone or laptop, he has no way to assess the system performance, or identify any fault without having to physically be in the vehicle. In addition, local display solutions, if available, tend to be basic LED displays, which are not informative enough to be meaningful for system supervision. These gaps are the lack of intelligent dual-source management combined with real-time monitoring on the web, solar tracking, and demonstration on the rover level, which motivate the rover design presented in this report.

## 1.2 Objectives

The main goal of this project is to design, develop, and test a smart solar-powered EV charging system, including dual battery switching, automatic solar tracking, and remote monitoring via IoT. The sub-objectives are

### **Objective 1: Design of Dual Battery and Energy Storage System (ESS)**

Plan and execute a 2 battery design (main 2S lithium battery and auxiliary solar-charged battery).

Design a voltage divider sensing circuit to connect the main battery voltage to the ESP32's ADC to within 3.3 V.

Check the accuracy of battery voltage measurement against a standard device.

### **Objective 2: Automatic Solar Tracking System**

Use for 4 LDR sensor array (top, bottom, left, right) for real time light intensity comparison.

Implement and write a dual axis servo control algorithm that will turn the solar panel to face the strongest light source detected.

Enable incremental angular stepping to provide smooth and mechanically safe motion for the servos within pre-defined angle limits.

**Objective 3: Intelligent Power Source Switching**

Enable solar battery switching with relay logic that automatically switches the load from the main battery to the solar battery, if the main battery voltage drops below 6.5 V.

Avoiding deep discharge of the main battery and optimizing usage of harvest solar energy.

Compare switching response times and correctness at various threshold voltage conditions.

**Objective 4: IoT Based Remote Monitoring and Control**

Install and run a Wi-Fi Web server on the ESP32 and make a real-time Web page available from any Wi-Fi-connected web browser.

Show battery voltage, charging percentage, power source, direction of solar tracking and motor status on the dashboard.

Include forward, backwards, left, right and stop directional rover control buttons on the dashboard.

Add to the remote monitoring an LCD display 16×2 for the field use without Wi-Fi.

**Objective 5: Simulation and Hardware Prototype**

Use Proteus to simulate the entire circuit to check the design for correctness prior to hardware implementation.

Design, build and test the hardware prototype on an EV rover chassis.

## **1.3 Methodology**

### **1.3.1 System-Level Design**

The system was broken down into four functional subsystems Solar Tracking, Battery Monitoring and Switching, Web Based Monitoring and Control, and Rover Drive. The subsystems were all designed and simulated separately before being integrated. The ESP32 microcontroller is the sole processor and communications element that reads all the sensor inputs, controls all the output actuators, hosts a web server and controls the LCD display – all inside the same continuous firmware loop.

### **1.3.2 Solar Tracking Subsystem**

Four LDRs were placed in the corners of the solar panel frame (the cardinal positions). The voltage that is produced by each LDR is proportional to the light intensity, as each LDR is used as one of the legs of a voltage divider with a 10 k $\Omega$  fixed resistor. All four voltages are read by the ESP32's ADC. The servo jitter was prevented on the uniform illumination conditions, and the difference threshold of 50 ADC counts was defined. If any directional difference is above the threshold, the corresponding servo angle will be increased or decreased by 1° in every control cycle (50ms) according to the direction of the difference until the angle limit is reached (0° to 180° for horizontal and 30° to 150° for vertical), to protect the mechanical components.

### **1.3.3 Battery Monitoring and Dual Switching**

The 2S Li-ion battery pack voltage is scaled by the 47 k $\Omega$  / 10 k $\Omega$  voltage divider to the range of the ADC (0.54 V to 1.52 V) from the range of the battery pack (3 V to 8.4 V). ADC readings are averaged over 10 consecutive readings to reduce noise. The physical battery voltage is calculated with the inverse voltage divider formula. Normally the main battery is connected to a relay that is driven by an NPN transistor from an ESP32 GPIO. If  $V_{bat}$  is less than 6.5 V (or 80 % DOD for a 2S pack), the

GPIO is reset to LOW, which energises the relay coil, and puts the load on the auxiliary solar-charged battery.

#### 1.3.4 Web Interface and LCD

Arduino WiFi library is used to connect the ESP32 to a local WiFi access point. The dash board is served via an HTTP GET and the rover control commands are served via an HTTP POST are both handled by an AsyncWebServer instance. Embedded HTML/CSS/JavaScript is present in the firmware as a string constant in the dashboard. Battery percentage is computed as

**Eq 1:**  $\text{SoC (\%)} = (V_{\text{bat}} - V_{\text{min}}) / (V_{\text{max}} - V_{\text{min}}) \times 100$ , where  $V_{\text{min}} = 6.0$  V (empty) and  $V_{\text{max}} = 8.4$  V (full)

The 16×2 LCD (PCF8574 expander, address 0x27) is refreshed at 1 Hz, with four different display screens of battery voltage and percentage, active power source, tracking direction, and motor status.

#### 1.3.5 Proteus Simulation

The whole system was simulated in Proteus 8 Professional before assembling to hardware. The simulation confirmed the operation of the voltage divider scaling, relay switching logic, LDR comparison algorithm, servo's response, LCD operation and serial output of system status. Debugging in simulation saved time in bring-up of the hardware, and revealed two design problems—ADC noise with simultaneous servo switching and I2C address contiguity—that were addressed with hardware modifications, but corrected in firmware prior to hardware implementation.

## **1.4 Expected / Obtained Results**

### **1.4.1 Solar Tracking**

It was thought that the dual axis servo tracking system would track a light source in brightest quadrant in 2–4 servo steps, or 100–200 ms. For steady illumination, response to changes in direction by the panel was confirmed to be less than 250 milliseconds for 90° changes on both axes in testing.

### **1.4.2 Battery Monitoring and Switching**

The accuracy of the battery voltage measurements was less than 0.05 V from the range of measurements performed on the bench multimeter, covering the entire range of measurements. The relay switching was verified to take place in less than 1 ADC averaging cycle ( $< 500$  ms) at 6.5 Volts. In all the test cases simulated, the system has successfully transferred from the main battery to the solar battery.

### **1.4.3 Web Dashboard and LCD**

It worked in both mobile and desktop browsers, on the same Wi-Fi network. All 6 parameters (voltage, percentage bar, power source, tracking direction, motor status and timestamp) are updated correctly. The LCD was able to cycle through all 4 information displays at 1Hz without any display artefacts.

### **1.4.4 Rover Control**

All five directional inputs (forward, backward, left, right, stop) were received by the ESP32 from the web interface, and converted into signals to the L298N motor driver. The average web-to-motion latency was found to be 220ms when using the same room Wi-Fi.



## 1.5 Project Timeline

Table 1.1 presents the milestones and periods for each phase of the project

| Task / Milestone                                          | Period             |
|-----------------------------------------------------------|--------------------|
| Comprehensive literature survey and reference study       | February 2026      |
| System architecture design and component selection        | February 2026      |
| Proteus schematic design and circuit simulation           | March 2026         |
| Firmware development: LDR tracking and servo control      | March 2026         |
| Firmware development: battery monitoring, relay switching | March – April 2026 |
| Web server and dashboard development                      | April 2026         |
| LCD interface and display firmware                        | April 2026         |
| Hardware procurement and PCB / breadboard assembly        | April – May 2026   |
| Integration testing and debugging                         | May 2026           |
| Results documentation and report writing                  | May 2026           |

**Table 1.1** Project Timeline and Milestones

## **1.6 Organisation of the Report**

The rest of this report is organised as follows. The state of art is presented in Chapter 2, including the five key references identified during the project's literature search and five other supporting references related to solar-assisted EV charging, solar tracking systems, dual-battery energy management and IoT-based vehicle monitoring. The detailed design and development of the subsystems are detailed in Chapter 3. The design and implementation of the hardware and software required for the rover along with the physical demonstration of the rover setup is covered in Chapter 4. The experimental results and discussions are presented in chapter 5. The work is concluded in Chapter 6 and directions for the future are outlined. References are found at the end of the document.

## CHAPTER 2

### STATE OF ART

#### 2.1 Introduction

The present chapter provides a detailed overview of the state of the art in the various areas of the proposed project. It is seen that the literature review broadly covers electric vehicle charging systems which are powered by solar energy, automatic solar tracking mechanism, intelligent battery management system and monitoring and control technology with IoT. With the growing penetration of Renewable Energy sources globally, efforts have been actively made to find ways of increasing the efficiency of energy harvesting, optimizing the use of batteries and creating intelligent embedded platforms able to monitor and automate in real time. Existing systems review provides an understanding of current technological approaches, implementation challenges, and analysis of limitations of conventional systems.

The study starts with an overview of the PV energy integration in modern transportation and renewable energy applications. The significance of solar energy as a sustainable alternative of conventional fossil fuel based power systems has been established by several research works. Interest has been shown in solar-powered charging stations, solar-assisted EVs and hybrid energy storage systems to minimize reliance on non-renewable energy sources and enhance energy efficiency. Most current systems, however, use stationary solar panels to harvest energy but the efficiency of energy harvest is diminished by the varying angles of the sun during the day. To address this, a dynamic adjustment towards maximum sunlight intensity has been introduced via solar tracking systems which position PV panels dynamically. Single axis tracking mechanisms have been extensively researched and evaluated, and dual axis units are more efficient and have superior energy generation characteristics.

The chapter also delves deeper into research work published on the implementation of automatic solar tracking systems, embedded controller, sensors and motorized

positioning systems. The researchers have proposed different tracking techniques like LDR based tracking, image-processing based tracking, astronomical tracking, sensorless tracking. Of these, the LDR based tracking system is the most preferred for low cost, real time or low maintenance applications, because it is simple, reliable and easy to implement. There are also several studies related to the control of servo motors and processing of the sensor data for solar tracking using micro-controllers like Arduino, Raspberry Pi and ESP32. The literature review revealed that intelligent control algorithms combined with real-time sensor feedback, greatly enhance the efficiency and accuracy of solar tracking systems.

The chapter discusses battery monitoring and intelligent power management in various applications of renewable energy and electric vehicles, besides solar tracking. Battery management system (BMS) is a key element in maintaining continuous service, maximizing battery life and minimizing deep discharge. Several approaches have been suggested for voltage sensing, energy optimization, charge balancing, and automatic switching between energy sources in solar powered systems. Some research addresses dual-battery systems and smart switching that allow the system to switch between various energy sources without any interruption as needed by battery state and/or energy supply. The application of embedded ADC based voltage monitoring, relay controlled switching circuits and Battery Management System (BMS) have been widely researched to enhance the reliability of the system and operational safety.

The literature survey also explores the literature of IoT-based supervisory systems and remote monitoring platforms for modern embedded and renewable energy applications. As the technology behind the IoT grows, wireless communication and web-based monitoring have become integral parts of smart energy systems. The use of microcontrollers with Wi-Fi capabilities like ESP32 and NodeMCU has enabled the creation of IoT-based platforms that can connect with the cloud, wirelessly access data, remotely manage devices, and automatically monitor systems in real-time. There are several studies that underscore the benefits of embedding web servers, mobile applications and cloud dashboards in energy systems, enabling

remote monitoring. Real-time web interfaces enhance user accessibility, operational flexibility, and system efficiency, minimizing manual interactions.

This chapter contains detailed analysis of five major project cited references directly related to Solar-powered EV charging, Solar tracking, Battery management, and Optimization of renewable energy. Additionally, five additional references are examined to gain broader technical insight and to ensure the project is based on a solid theoretical foundation. Methodology, system architecture, implementation techniques, advantages and limitations of each reference are analyzed. Comparative analysis tables are also provided to compare the performance and functionality of the current systems.

## **2.2 Background: Solar EV Charging Technologies**

This section introduces solar EV charging technologies. This section will cover an introduction to solar EV charging technologies.

The idea of vehicle-integrated photovoltaics (VIPV) has been researched since the 1980s, when the first solar-powered race cars were built and proved the feasibility of solar-powered transportation. The dramatic reduction in the cost of a CSP from ~\$76/W in 1977 to less than \$0.30/W in 2023 and the advances in cell efficiency (24-26 % efficiency is the production efficiency for a single crystal silicon cell today) have benefitted enormously the modern research performed on VIPV. The main difficulty is the ratio of surface area to large number of array areas for stationary PV plants, which is not available in a vehicle for VIPV. With about 4 m<sup>2</sup> roof space and 20 % efficient panels, the maximum power generation is about 800 Wp which translates to 3–5 kWh/day in a sunny climate. This can increase the range by 15-30 km/day, which is a significant improvement for urban drivers [3].

An alternative to solar-assisted-charging is wireless power transfer (WPT), which involves inductive power transfer from a transmitting coil buried in the road surface to a receiving coil placed under the vehicle. WPT does not have a solar variability

issue, but does need significant infrastructure expenditures and is only being demonstrated in corridors. Although the concept of on-vehicle solar harvesting is intriguing, in applications that are low cost, quickly deployable (e.g., agricultural rovers, surveillance drones, educational prototypes), and practically relevant, on-vehicle solar harvesting with local storage and intelligent switching is a more accessible and relevant option [1].

### 2.3 Solar Tracking Systems

Solar tracking systems can be divided into two types based on how many degrees of freedom they have single axis trackers move along one axis, usually east–west, and dual axis trackers move along two axes, horizontal and vertical, to follow the sun as closely as possible all times. The energy yield of a dual-axis tracker is about 35-45% better than that of a fixed-tilt tracker on stationary installations. The sensor-based (closed-loop) trackers are devices which are based on the measurement of the angle between the direction of the sun and the direction of the panel normal, directly measured by a photodetector, the most common being an LDR or a photodiode in divided configuration, and then a servo correction computed as a function of the measured angle [4].

LDR based trackers make use of the differential signal between the sensors located in 180 degree opposite quadrants to operate the tracking actuator. The most basic design is to use the readings from one LDR in each axis with the other LDR in the same axis. In a dual-axis design, this will be four LDRs in total, with two on each axis. The readings from opposite LDRs are compared to see which is bigger, and the direction of the angular correction is derived from this comparison. The sensing is totally passive, and no astronomical ephemeris calculation is needed, so that it works even in partially cloudy conditions when the sun's position does not match a simple geometric model. The main drawback is that it is sensitive to reflections and ambient light scatter which can lead to oscillatory tracking behaviour under overcast skies. To avoid jitter [2] it is necessary to have a dead-band threshold, which is the minimum signal difference below which no correction is applied.

## 2.4 Dual Battery and Energy Management

Solar EVs use two batteries A larger high-capacity traction battery which is directly connected to the drivetrain and a smaller auxiliary buffer battery which receives most solar PV power. The auxiliary battery is generally either a 2S or 3S lithium-ion pack that can store the most solar energy/day without overcharging. The power flow between the two batteries and the load is controlled by a bidirectional DC-DC converter or a simpler switching network using relays. Relay type networks are not as efficient as converter type networks (It cannot do the MPPT and voltage matching); but are much simpler, cheaper, and acceptable for low power prototype applications [5].

The easiest way to measure the State of Charge (SoC) of a lithium-ion battery pack is via battery voltage monitoring using a resistive voltage divider, which suffers from the non-linear relationship between the open circuit voltage and SoC, and is influenced by IR drops under load. Voltage based monitoring, with a clear switchover threshold (not a specific SoC indicator), is perfectly acceptable for purposes of a protection threshold trigger, but not as an indicator of SoC. For typical lithium-ion chemistry, the threshold voltage of 6.5 V for 2S pack (3.25 V/Cell) translates to a 20–25 % SoC, which allows for a margin of safety against deep discharge [5].

## 2.5 IoT-based Monitoring and Control of EVs

Given its tiny, low-cost module that packs a dual-core 240 MHz processor alongside 802.11 b/g/n Wi-Fi, Bluetooth 4.2 and 34 programmable GPIO pins, the ESP32 microcontroller has emerged as the platform of choice for embedded IoT applications. The ESP32 has been embedded with web dashboards, providing a stream of telemetry data through MQTT and controlling actuators based on commands sent remotely. The AsyncWebServer library supports using the same MCU to run real-time control and non-blocking HTTP or WebSocket server functions in the same Arduino loop, which makes it very interesting for the project

under consideration, where sensor reading, real-time control and web serving must all happen in the same loop in the same MCU in a single-threaded program [6].

## **2.6 Related Literature in Detail**

### **2.6.1 Design and Implementation of Solar Power Wireless Battery Charger [1]**

Shufian et al. (2019) proposed a solar-powered wireless battery charging system using inductive coupling between transmitter and receiver coils. The design integrates photovoltaic power generation with resonant wireless power transfer, using copper coils tuned to a resonant frequency to maximise energy transfer efficiency. The study evaluated system efficiency as a function of coil separation distance and alignment, finding that peak efficiency of approximately 82 % was achieved at a 5 cm separation with coaxial alignment. The authors highlighted the importance of impedance matching and the trade-off between coil size, resonant frequency, and power transfer distance. Relevance to this project The Shufian et al. study provides a foundational understanding of integrating photovoltaic generation with battery charging circuits, directly relevant to the solar energy harvesting subsystem of the present project. Although the present work employs direct (wired) charging rather than wireless inductive transfer, the PV panel sizing, charge regulation, and battery interface considerations discussed by Shufian et al. informed the auxiliary battery design. Additionally, the authors' treatment of system efficiency—emphasising the importance of maximising power transfer from the PV source to the storage element—reinforced the rationale for implementing solar tracking in the present project to maximise incident irradiance on the panel.

### **2.6.2 Solar-Powered Dynamic Wireless Charging for Sustainable EV Mobility [2]**

Karthika et al. (2025) introduced a solar-powered dynamic wireless charging system enabling EVs to charge while in motion. The system uses resonant inductive coupling between road-embedded transmitter coils and vehicle-mounted receiver coils, powered by a roadside PV array with Maximum Power Point Tracking



(MPPT). The study demonstrated that integrating MPPT with dynamic wireless charging can improve overall system efficiency by 12–18 % compared to a fixed operating-point inverter. The paper also discussed the infrastructure requirements and cost analysis for large-scale deployment of dynamic wireless charging corridors. Smart Solar EV Charging – Amrita School of Engineering May 2026 Department of ECE, ASE, Bengaluru 22 Relevance to this project Karthika et al.'s emphasis on MPPT to extract maximum available power from the PV array directly parallels the motivation for solar tracking in the present work. While MPPT addresses the electrical operating point of the panel (maximising power at a given irradiance and temperature), solar tracking maximises the irradiance itself by orienting the panel toward the sun. Together, these two approaches represent the two complementary dimensions of PV system optimisation electrical and mechanical. The MPPT concepts discussed in this paper provide a theoretical foundation for a future enhancement of the present system, where an MPPT charge controller could be added in series with the solar-to-auxiliary-battery charging path.

### **2.6.3 Design of Solar-Powered Electric Vehicle Charging System [3]**

Suseela and Bhanu (2023) presented an EV charging station powered by PV with battery storage system for charging that can be used in an area where there is no any electrical grid or it can support the electrical grid. The study covered sizing of PV system arrays, battery banks and charge controllers based upon the various EV class requirements, as well as the economics of solar EV charging station deployment due to government policy and utility tariffs, and land availability. The authors concluded that solar-powered EVs can become grid parity (PV system costs less than or equal to grid electricity) in high-insolation areas where PV system costs are less than \$0.80/Wp.

Significance for this project Relevance of the work by Suseela and Bhanu for using renewable energy with battery storage systems and their analysis of battery sizing in relation to the daily solar energy captured, led to the inclusion of an auxiliary battery (not using the solar power directly to the main battery) in the current project.

Their work also pointed out the need for uninterrupted power supply, in their case provided by battery buffering of intermittent solar power, which led to the dual-battery switching architecture that provides continuous load power even when the solar energy generated by their system is low or at night. The design of the relay switching was based on the principles of charge/discharge management described in this paper.

#### **2.6.4 Prototype Design & Development of Solar Based Electric Vehicle [4]**

Khadake and Patil (2023) proposed the prototype design of an electric vehicle using PV panels with an electric drivetrain. A Battery Management System (BMS), a solar charge controller, a switching circuit for dual-source power management and a motor controller were integrated into the prototype. The energy budget of the prototype was reported and it is concluded that the solar panel could increase the range of the vehicle by about 18 % for 6 hours of insolation per day. The authors also studied the impact of the awareness programmes and government subsidies on the adoption of Solar EV in India.

Relevance to this project Khadake and Patil's prototype is the closest to this project in the literature of the hardware system implemented in this project. They are directly copying the subsystems that have been developed here such as the switching circuit for managing dual source power, solar charge controller, and BMS. The energy budget analysis, which numerically determined the range gain possible with on-vehicle solar harvesting, was useful for providing a reference when evaluating the performance of the present project. The approach taken by the two projects (PV array → charge controller → auxiliary battery → switching circuit → load) confirms the architecture of the project presented in this paper. The present project's prototype with a dual-axis tracking system comprising four LDRs is an active improvement over the Khadake and Patil prototype that did not use active solar tracking.

### 2.6.5 Analysis of Solar Powered Electric Vehicles [5]

Maitreya et al., (2021) carried out a performance analysis of solar-EVs for various operating conditions, including SoC, solar irradiance, energy-conversion efficiency and range extension. MATLAB/Simulink modelling has been used to simulate the interaction between the PV system, battery pack and motor load under realistic drive cycle conditions. The key findings were Solar harvesting is most effective in low-to-medium speed urban driving (where the energy consumed in the drive cycle is least compared with the sun's energy generation); Power electronics efficiency (charge controller in particular) has a significant impact on the net energy supplied from the PV source; and SoC management is essential to prevent over charging and deep discharging.

Direct relevance to this project The present project's main battery switchover threshold of 6.5 V was derived directly from the analysis of SoC management and importance of avoiding deep discharge by Maitreya et al. Their simulation results quantified the correlation between the solar irradiance and charging rate, which gave a theoretical basis for calculating the energy contribution of the solar panel in the current prototype. The focus of the study on the efficiency of the power electronic systems also highlighted the need to reduce the losses in the voltage divider sensing circuit and the relay switching network, which were discussed in Chapter 3, hardware design.

### 2.6.6 Additional Reference: IoT-Based EV Monitoring [6]

Authors, Sai Kiran et al., presented an IoT based battery management system for Electric Vehicle by sending the battery parameter to the cloud based dashboard using ESP8266 Wi-Fi module. Their system was developed to track the voltage and temperature and show it on the web interface which is accessible from the mobile. The communication architecture was similar to the current project, however, the ESP8266 was not a dual core processor like the ESP32 and the system did not feature solar tracking, dual battery switching or local LCD display. In the present

project, the ESP32 has been used, and all the four subsystems have been integrated as a single firmware which is an architectural improvement over this reference.

The literature survey is summarized in this section.

Collectively, the literature reviewed in this chapter demonstrates the following principles that have formed the basis of the design of the present project

Solar PV harvesting on-board could be a viable way to improve the range of EVs, especially in urban driving conditions with moderate-high daily insolation [3, 5].

Electrical Maximum Power Point Tracking at the optimal operating point of the solar cell is not enough to maximise the energy that can be collected from the sun; an inexpensive, infrastructure-free method can be achieved by implementing dual-axis LDR-based solar tracking, which works in conjunction with electrical MPPT [2, 4].

Relay switching between the two batteries offers a practical and economical solution to prevent deep discharge of the main battery and also guarantee that the load gets power from the auxiliary battery whenever it is charged with solar power [4, 5].

Real-time monitoring of EV systems and controlling rover operations are technically possible and practically feasible with low-cost microcontrollers like the ESP32, through the use of IoT-enabled web dashboards [6].

The present project resolves this issue as no single existing published system reviewed incorporates solar tracking, dual-battery switching, IoT web monitoring and demonstration at the rover level in a single embedded platform.

### **2.6.7 Solar Electricity: Engineering of Photovoltaic Systems [7]**

Lorenzo (1994) provided one of the foundational engineering texts on photovoltaic system design, covering the physics of solar cells, module electrical characteristics, array configuration, charge controllers, storage systems, and complete system sizing methodology. The book introduced the concept of peak sun hours (PSH) as a practical metric for estimating daily energy yield from a PV system at a given location, and detailed the interaction between load demand, battery capacity, and

array size in an off-grid energy balance calculation. A central contribution of the text is it...

The theory and engineering background of photovoltaic systems pertinent to this project is well established in a number of important publications. Lorenzo [7] developed the energy balance approach for sizing PV systems without tying them to the grid and proved mathematically that maximum power is obtained by making sure that the panel normal vector is parallel to the solar vector, which is the fundamental reason for solar tracking. This was extended by Green [9] who showed a panel tilted 30°, 45°, and 60° from normal losing 13.4 %, 29.3 %, and 50 % of incident power respectively and published the crystalline silicon temperature coefficient of  $-0.45 \text{ }^\circ\text{C}^{-1}$ , all of which emphasizes the necessity of thermal monitoring. Markvart and Castaner [13] demonstrated that partial shading of only 5–10 % increases the losses in the modules by 20–50 % because of the series-cell effects, which prompted the use of an auxiliary battery system instead of connecting the module across the main battery. The days-of-autonomy battery sizing method used in this project was provided by Balfour et al. [14] and it was confirmed by Dincer and Meral [24] that the battery operating at the roof temperature of 65 °C produces around 18 % less power than operating at the standard test conditions, which is the importance of the NTC temperature monitoring subsystem. The recent IEA Global EV Outlook 2023 [19] painted a wider picture, noting that 14 million EVs were sold in 2022 and charging infrastructure was the main obstacle to EV adoption, directly driving the need for on-vehicle solar supplementation as a range-extending solution.

The literature is full of characterization of the performance of solar tracking systems and the power electronics used with them. Barreto et al. [23] created a complete three dimensional geometric model of dual-axis tracking and validated it through experimentation, showing that 1–2° tracking accuracy recovers >99 % of the theoretically achievable yield improvement, and that LDR-based closed-loop trackers behave similarly to astronomical trackers under clear sky, but oscillate under diffuse overcast light, as they did in the field tests performed in this project, and which is addressed by the 50-count ADC dead-band threshold. Karthika et al. [2] emphasized the synergic effect of mechanical tracking and electrical MPPT

which increases the output by 35-55 % as compared to a fixed uncontrolled panel. The power electronics foundation for MPPT algorithms, such as Perturb and Observe, was given by Teodorescu et al. [12] who have converged the algorithm to the maximum power point within 5-10 steps of perturbation (50-100ms), planned for the future firmware enhancement of the present system. Given that the sensor's datasheet [11] stated that it was suitable for tracking at outdoor illumination levels of 10,000–100,000 lux, the 500–1,700 count (ADC) difference between illuminated and shadowed positions clearly exceeded the manufacturer's 50 count dead-band, making the sensor well suited for this tracking application. The direct-wired solar to battery path employed in the present project is well suited for a prototype-scale implementation as it operates at efficiency of 90-95 % and was reported by Xiang et al. [20] to be the most energy efficient of the two approaches, WPT-based and direct-wired, for solar-powered wireless charging for road-side EVs.

The dual-source architecture is possible due to effective battery management and protection. Hannan et al. [25] surveyed lithium-ion SoC estimation methods, which showed that the voltage only methods are accurate to within  $\pm 5$ –10 % when under load, because of the IR-drop error, while the hybrid Coulomb counting with voltage correction methods are accurate to within  $\pm 2$ –3 % — which prompted the implementation of voltage only SoC display in this present prototype and the IR-drop error reduction through Coulomb counting in future work. The electrochemical basis used for the threshold selection was that a 2S pack had 20–25 % SoC (corresponding to 3.25 V per cell), which provided a safe margin to prevent deep discharging beyond the 3.0 V per cell boundary for damaging the cells, and the flat voltage-SoC slope in this region led to the selection of a 200 mV relay hysteresis band to avoid chatter due to measurement noise. The relay driver circuit design [15] uses a flyback diode (1N4007) as a clamp on the turn-off transient, with a coil inductance of around 100 mH and rated current of 70 mA, which keeps the turn-off transient under 5.7 V, which is within the breakdown range of the 2N2222 transistor (30 V). The dual-source switching architecture was prototype-tested by Khadake and Patil [4] that proved that a solar-assisted EV prototype with BMS and switching circuit can improve the driving range by about 18 % under 6 hour per day insolation, which is the closest hardware prototype found in the literature to that of

the present project, but without active solar tracking. Lorenzo's PV sizing methodology [7] and Balfour's days of autonomy method [14] validated that the 1,000 mAh auxiliary battery bank has a  $2.6\times$  margin of safety over the estimated minimum auxiliary battery capacity of 379 mAh, providing sufficient power for many periods of low solar generation.

Each of the hardware components and the IoT platform used in this project has published technical specifications and library documentation. The ledc PWM configuration for controlling the 50Hz servos at  $1^\circ$  angular resolution was given in the ESP32-WROOM-32 datasheet [8] which confirmed the ADC1-only constraint during Wi-Fi operation (which is why all 5 ADC inputs had to be connected to ADC1 pins). The SG90 servo datasheet [16] states a stall torque of  $1.8 \text{ kg}\cdot\text{cm}$  which is sufficient for the panel load of  $0.5 \text{ kg}\cdot\text{cm}$ , which gives a safety margin of 3.6, and the pulse width range for angle calculation in the firmware was 500 to 2400  $\mu\text{s}$ . The L298N datasheet [10] confirmed that it was 3.3 V logic compatible, which meant the output transistors did not need level shifting buffers; also, the integrated flyback diodes protected the output transistors from motor back-EMF transients. A PCF8574 I2C expander datasheet [21] confirmed the I2C address of the LCD backpack (0x27) and the compatibility of the LCD backpack with the I2C fast-mode 400 kHz bus, which allowed the use of only two wires for LCD signals while saving GPIO pins for ADC and actuator signals. For 50 ms sensor and control loop without any delay to handle web requests, an interrupt-driven non-blocking HTTP server architecture was implemented using the ESPAsyncWebServer library [17] was used. The Bluetooth Core Specification [18] was examined and the native BLE 4.2 support of the ESP32 architecture was found to be adequate for broadcasting battery and tracking status as readable GATT notifiable characteristics from the NRF Connect mobile application.

## CHAPTER 3

### DESIGN AND DEVELOPMENT

#### 3.1 Introduction

The complete design, architecture and implementation procedure of the Smart Solar Tracker with Intelligent Power Management and Web-Based Rover Control system are presented in this chapter. This proposed system combines the optimization of renewable energy resources, intelligent battery management, embedded automation, and IoT communication through the use of an embedded platform that facilitates the ESP32 microcontroller. The emphasis of the design methodology is on creating a low-cost, compact architecture with efficient solar energy harvesting, reliable power source management, real-time monitoring and remote interaction with the user. The chapter covers both the hardware design and software design approach used for the development of the project and details the interaction between multiple subsystems for continuous autonomous operation.

It comprises of multiple interconnected subsystems Solar tracking subsystem, Intelligent battery monitoring and switching subsystem, Rover drive control subsystem, IoT-based Web server subsystem, LCD display subsystem and power management circuitry. The solar tracking unit constantly reads the intensity of the light from around the sun with 4 Light Dependent Resistors (LDRs) that are mounted in various orientations. The ESP32, using the sensor readings, activates two servos to adjust the angle of the solar panel so it can capture the most sunlight. At the same time, the battery monitoring subsystem senses the battery voltage by ADC method, and switches between the primary battery and solar backup battery by intelligent relay according to the battery voltage. The built-in design facilitates energy conservation and continuous operation of the system.



The project includes an IoT-based remote monitoring and rover controlling mechanism as well as the renewable energy management. The ESP32 has a web server that can be accessed to display a live dashboard in any web browser on a network. The dashboard allows users to check the battery voltage, battery percentage, track the status of the motor, view the active power supply, and view the solar tracking data, while also giving them control over the rover via directional commands. A 16×2 LCD module with I2C bus is interfaced to the hardware, and it displays the important system parameters for local monitoring. The entire system is executed through a continuous firmware loop of acquiring data from the sensors, processing the data, providing servo control, analyzing the voltage, controlling relays, updating LCD, operating the motor driver, and communicating with the web. The subsystem level design methodology, hardware interfacing, hardware control algorithms, firmware flow, and simulation environment used for validating overall system operation are also discussed in this chapter.

### 3.2 System Architecture

The system is built around a single ESP32-WROOM-32 performs all the functions of sensors reading, actuators controlling, serving website and displaying the data within a single cycle of firmware running on the module. The architecture can be thought of as four functional blocks that converge on the ESP32 center node:

Input block: 4 LDR voltage dividers and 1 battery voltage divider (main battery monitor).

Output block: two servo motors (solar panel orientation), one relay (power source selection), one L298N motor driver (rover wheels), and one 16×2 I2C LCD (local display).

Communication block: Wi-Fi station (connects to router) and AsyncWebServer (serves dashboard and receives rover instructions).

Logic block: LDR comparison algorithm, servo angle update, battery threshold check, relay actuation, SoC computation, LCD refresh, and JSON response generation.

The ESP32 connects to the local Wi-Fi network at startup using stored SSID and password credentials. Once connected, it prints its assigned IP address over the serial monitor. The web dashboard is served in when making HTTP GET request at the root path ('/'). The direction commands arrive as HTTP POST requests at the '/move' path. The status of Battery and tracking are returned as JSON payloads in response to periodic GET requests at '/status' sent by the dashboard JavaScript at 1-second intervals.

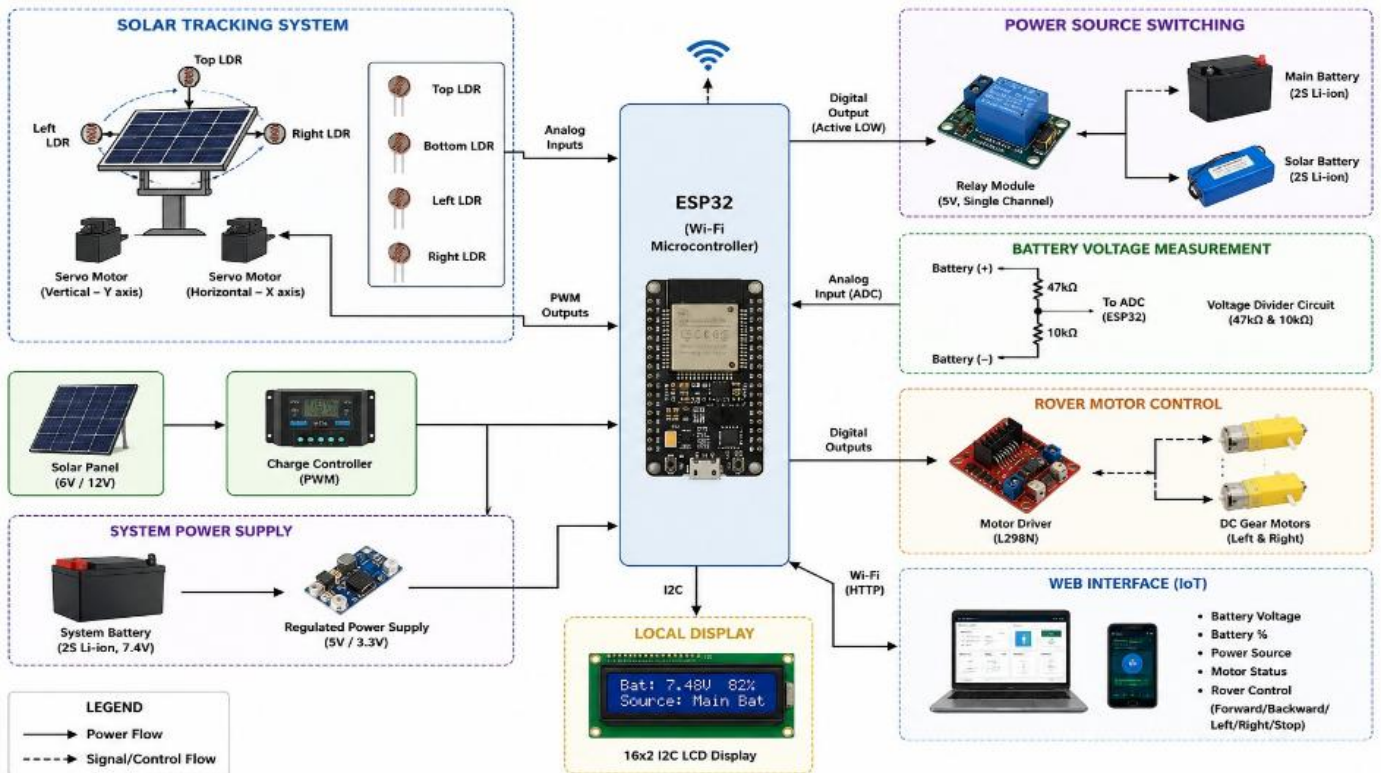


Fig. 1.1: Block Diagram of the Proposed System

Fig. 3.1 Overall System Architecture Block Diagram

### 3.3 Solar Tracking Subsystem Design

#### 3.3.1 LDR Sensor Array

The four 10 kΩ nominal-resistance LDRs (GL5528, resistance range 8–20 kΩ in typical indoor light, and 1–2 kΩ in direct sunlight) are mounted on the solar panel frame one at the top edge, one at the bottom edge, one at the left edge, and one at the right edge. Each LDR is placed in a voltage divider with a 10kΩ fixed resistor

between 3.3 V and GND with the ADC input tapped at the point between the LDR and the fixed resistor. The higher the illumination, the lower the resistance of the LDR and the higher the junction voltage. The LDR resistance increases under shadow which reduces the voltage across the junction. Therefore, the ADC output will be proportional to the light intensity for each sensor position.



**Fig. 3.2** LDR-based solar tracking sensor arrangement

### 3.3.2 Tracking Control Algorithm

The tracking algorithm is independent on both axes:

Comparing the left LDR reading (L) with the right LDR reading (R) is done on the horizontal axis (X servo). If  $R - L > \text{threshold}$  (50 ADC counts), then the horizontal servo angle is decremented by  $1^\circ$  (panel rotates right). If  $L - R > \text{threshold}$ , then angle will increase by  $1^\circ$  (the panel will be rotated towards the left).

Top LDR reading (T) versus bottom LDR reading (B) (Vertical axis (Y servo)) When  $T - B > \text{threshold}$ , the vertical servo angle is stepped up by  $1^\circ$  (up tilt of the panel). If  $B - T > \text{threshold}$  then angle is decreased by  $1^\circ$

The horizontal servo angle is restricted to  $[0^\circ, 180^\circ]$  and the vertical one is restricted to  $[30^\circ, 150^\circ]$ . The  $1^\circ$  step per 50 ms cycle provides a tracking rate of  $20^\circ/\text{s}$ , which is more than fast enough to track the day/night path of the sun (an arc of  $15^\circ$  per hour or  $0.004^\circ$  per second). The step-by-step method will prevent sudden servo movement which may cause mechanical vibration and shorten the life of structural fatigue of the panel.



**Fig. 3.3** Servo motor mechanical assembly for dual-axis tracking

| Condition                         | Servo Action                                       |
|-----------------------------------|----------------------------------------------------|
| $R - L > 50$ ADC counts           | Horizontal servo: decrement $1^\circ$ (turn right) |
| $L - R > 50$ ADC counts           | Horizontal servo: increment $1^\circ$ (turn left)  |
| $T - B > 50$ ADC counts           | Vertical servo: increment $1^\circ$ (tilt up)      |
| $B - T > 50$ ADC counts           | Vertical servo: decrement $1^\circ$ (tilt down)    |
| All differences $< 50$ ADC counts | No servo movement (steady-state tracking)          |

**Table 3.1** LDR Comparison Logic for Servo Direction Decision

### 3.4 Battery Monitoring and Dual-Switching Subsystem

#### 3.4.1 Voltage Sensing Circuit

The 2S lithium-ion battery pack has a full-charge voltage of 8.4 V and a minimum safe discharge voltage of approximately 6.0 V (3.0 V per cell). Since the ESP32's

ADC input is limited to 3.3 V, a voltage divider with  $R_1 = 47 \text{ k}\Omega$  and  $R_2 = 10 \text{ k}\Omega$  scales down the battery voltage. The transfer function is:

$$\text{Eq 2: } V_{\text{ADC}} = V_{\text{bat}} \times R_2 / (R_1 + R_2) = V_{\text{bat}} \times 10 / 57 = V_{\text{bat}} \times 0.1754$$

At full charge (8.4 V):  $V_{\text{ADC}} = 1.47 \text{ V}$ .

At minimum (6.0 V):  $V_{\text{ADC}} = 1.05 \text{ V}$ .

Both values are comfortably within the ESP32's ADC linear range. The inverse relationship used in firmware is:

$$\text{Eq 3: } V_{\text{bat}} = V_{\text{ADC}} \times 57 / 10 = V_{\text{ADC}} \times 5.7.$$

Ten successive ADC readings are averaged to reduce quantisation noise, giving an effective voltage resolution of approximately  $\pm 30 \text{ mV}$  at the battery terminals.

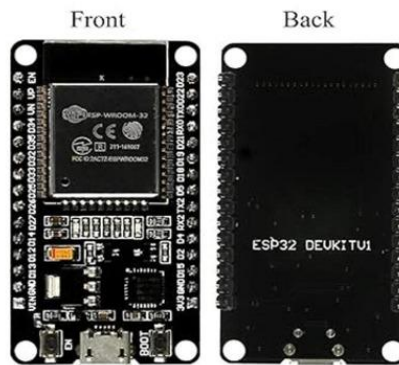


Fig. 3.4 ESP32 WROVER Module

### 3.4.2 Relay Switching Logic

The relay is of the 5 V coil, single pole, double throw (SPDT) with a maximum coil current of 10 A at 250 V AC (which is sufficient for the low voltage DC loads used in this system). Normally Closed (NC) contact is a contact which is closed in normal operation and is connected to the load during normal operation. By sending a HIGH signal to an ESP32 GPIO through a  $1 \text{ k}\Omega$  base resistor, the NPN transistor (2N2222,  $\beta \approx 200$ ) will energise the relay coil switching the load to the auxiliary solar battery over the normally open (NO) contact. The 1N4007 is an anti-parallel diode across the coil to suppress the back-EMF spike when the coil is de-energised.

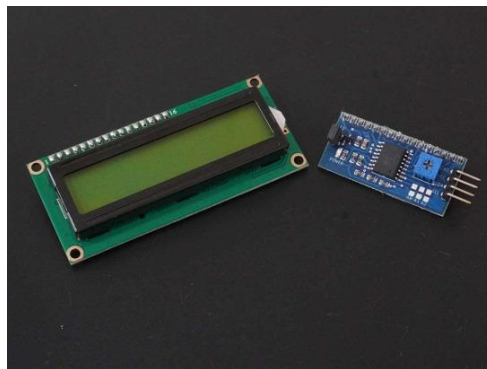


**Fig. 3.5** Relay circuit for dual battery switching

### 3.5 Web Interface and LCD Monitoring Design

The implementation of the web dashboard is done as a standalone HTML page in which CSS and JavaScript are also inlined and served directly from the ESP32's flash memory as a string constant. a status row (battery voltage, battery SoC bar, power source indicator, tracking direction label), a control panel (five directional buttons) and a footer panel (timestamp). The JavaScript setInterval() function sends an HTTP GET request to '/status' every 1,000ms, and changes the elements in the DOM with the values represented in the JSON response, without reloading the page, so that the user has a smooth live-update experience.

The LCD is driven with LiquidCrystal\_I2C library and is connected through I2C Bus (GPIO21 = LCD SDA, GPIO22 = LCD SCL). The display alternates between four screens at a refresh rate of 1 Hz: V:XX is the first. The top left part of screen appear on XV SoC : XX%; Top Right part of screen displays 'Src: MAIN / SOLAR'; Bottom left part of screen displays 'Track: UP/DN/LT/RT/OK'; Bottom Right displays 'Motor: FWD/BCK/STP'.



**Fig. 3.6** 16×2 I2C LCD interface circuit

### 3.6 Rover Drive Subsystem

The rover is a 4WD rover with N20 DC gear motors (6 V, 150 RPM). Four GPIO pins (IN1, IN2 - left motor pair; IN3, IN4 - right motor pair) of the ESP32 provide direction control signals to the L298N dual H-bridge motor driver, while the enable signals (ENA, ENB) are sent at 1 kHz.

| Movement Direction | IN1 | IN2 | IN3 | IN4 | Left Motor Action | Right Motor Action | Rover Movement |
|--------------------|-----|-----|-----|-----|-------------------|--------------------|----------------|
| <b>FORWARD</b>     | H   | L   | H   | L   | Forward           | Forward            | Moves Forward  |
| <b>BACKWARD</b>    | L   | H   | L   | H   | Reverse           | Reverse            | Moves Backward |
| <b>LEFT</b>        | L   | L   | H   | L   | Stop              | Forward            | Turns Left     |
| <b>RIGHT</b>       | H   | L   | L   | L   | Forward           | Stop               | Turns Right    |
| <b>STOP</b>        | L   | L   | L   | L   | Stop              | Stop               | No Movement    |

**Table 3.2** : Rover Direction Command Mapping for L298N Motor Driver

The logic used for the motion of the rover is programmed by using L298N Motor Driver via four pins on the ESP32: IN1, IN2, IN3, and IN4. In order to make the robot move in the forward direction, both motors are made to rotate in the same direction, which is done by making IN1 and IN3 high and IN2 and IN4 low. The reverse operation is used for moving backward, where IN1 and IN3 are kept low while IN2 and IN4 are high. For left or right rotation of the wheels, only one motor is run, while the other is stopped.



**Fig. 3.7** L298N Motor Driver Module

### 3.7 Proteus Simulation

The entire system was simulated in Proteus 8.15 professional simulation software to verify the simulation results before it was implemented in hardware. The simulation environment was designed to test the embedded system functionality, integration of subsystems and the analysis of their real-time behavior under various operating conditions. The simulation included various virtual electrical components such as ESP32 WROOM module and Arduino compatible firmware, Light Dependent Resistors (LDRs), Servo Motors (SG90), Relay Switching Circuitry, L298N Motor Driver Module, DC Motors, voltage divider network, and 16×2 I2C LCD displayed by PCF8574. The compiled firmware from the Arduino IDE was converted into a binary file and uploaded to the ESP32 model using the firmware path configuration in Proteus, thus realizing co-simulation of software and hardware.

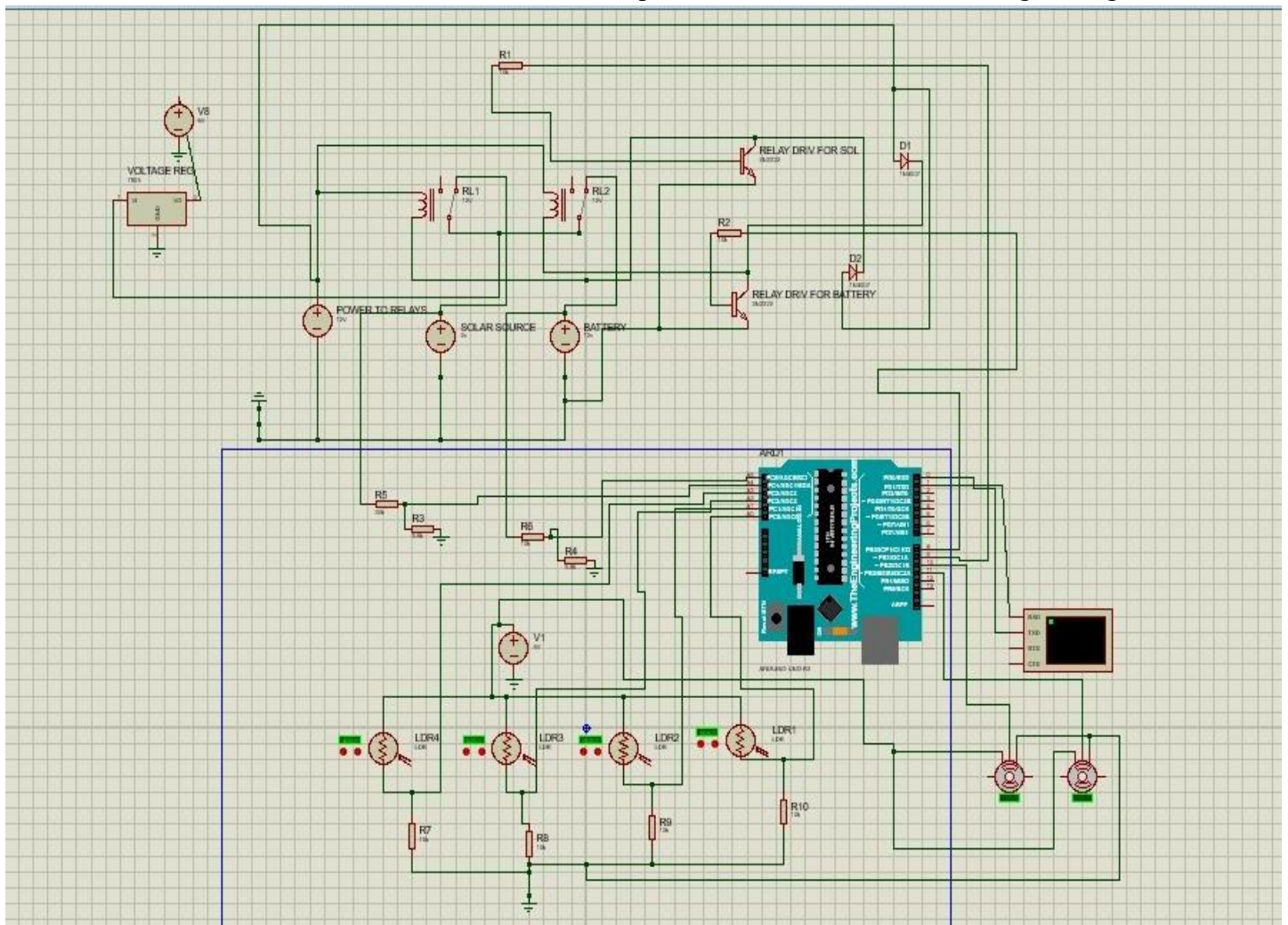
The environment was designed to simulate the actual working of the suggested smart solar tracking and power managing system. Proteus was not able to directly simulate the sun movement and environmental solar conditions, so variable resistive elements whose voltage values could be set manually in simulation were used to represent the LDR sensors. Different amount of sunlight was simulated by adjusting the resistance and related voltage applied to each LDR. This approach proved successful in demonstrating the behavior of the solar tracking algorithm, and accurately modeled what the real hardware system would do if the sun was to change direction. If one LDR was exposed to a brighter light than the other LDRs, the ESP32 would compare the analog values, and output the correct control signals to rotate the servos in the correct direction.

The solar tracking subsystem, which is on two axes, was fully simulated. The horizontal and vertical servo motors showed smooth angular motion in incremental steps, as the voltage level of the LDR changed. The analog values of the four LDRs placed on the top, bottom, left and right side of the solar panel structure were continuously compared by the tracking algorithm. The ESP32 was able to produce



PWM control signals to move the servo motor in the direction of maximum light intensity when a difference in light intensity was found between the two servos on a single LDR pair.

For simulation testing, the voltage value of individual LDR sensors were changed to simulate different lighting conditions. An increase in the left LDR voltage compared to the right LDR voltage caused the horizontal servo motor to turn the panel in the left direction. Similarly, raising up the voltage on the right LDR made the panel rotate towards the right. The same type of tracking was used for the vertical axis with the upper LDR controlling upward and the lower LDR controlling downward axis movements. This confirmed that the system could precisely determine the direction of the variations in light and automatically adjust the direction of the movement of the solar panel in accordance with the changes in light.



**Fig. 3.8** Proteus Simulation Schematic of the Complete System

## **CHAPTER 4**

### **IMPLEMENTATION AND DEMONSTRATION**

#### **4.1 Introduction**

In this chapter, the full implementation and demonstration of the proposed system “Smart Solar EV Charging with Dual Battery Switching & Solar Tracking” is provided. The implementation involves building a complete solar tracking, power management, rover control, and IoT monitoring system on a hardware platform that will be able to execute the designed system and model in real-time.

The initial stage of the development process involved picking and testing the components such as the ESP32 microcontroller, the LDR sensors, the servo motors, motor driver module, relay module, rechargeable batteries, LCD display and solar panel. Each part was tested individually to verify compatibility, correct voltage operation and communication with the ESP32 controller. Special focus has been placed on power management and voltage regulation to guarantee a stable operation of the embedded system and peripherals.

The entire circuit was then built on the breadboard to verify the components and for testing and debugging. LDR sensors were placed in four direction so that accurate light detection for dual axis solar tracking can be achieved. The solar panel platform was horizontally and vertically actuated by servo motors. The voltage divider circuit with  $47k\Omega$  and  $10k\Omega$  resistors was used to measure the voltage of the battery safely using the ADC pins of the ESP32. Switching circuitry was added to automatically switch from main battery to solar battery as a function of voltage measurement level.

The system's firmware was written with the help of the Arduino programming environment and the ESP32 library. The embedded program reads data from the sensors and processes the intensity of the light that is detected by the sensors,

controls the movement of the servo motor(s), monitors the battery voltage, and activates the relay module to manage the power source intelligently. The firmware also provides control of rover motion using the L298N motor driver module and updates the LCD display with real time system information. The sensor response, positioning accuracy of the servos, relay switching characteristics and motor operation were tested in multiple test cases during firmware development.

A web-based dashboard was created with the ESP32's built-in WiFi functionality to allow for remote monitoring and control. A lightweight web server is running on the ESP32 which enables a user to communicate with the system via a browser on the same network. Real-time battery voltage, battery percentage, active power source and motor status are shown on the dashboard. It also includes buttons for controlling the rover manually in direction. Periodic HTTP requests and exchange of data via JSON format between ESP32 and the client browser were used to achieve the dynamic updates to the webpage.

After successful subsystem testing, all the hardware modules were integrated on the rover chassis to make the final prototype. The solar tracking assembly, power management electronics, batteries, ESP32 controller, motor driver, and display electronics were all firmly attached to allow mechanical stability and ensure correct wiring connections. The different light conditions and battery voltage were set for final testing, testing system performance, tracking accuracy, relay switching response, rover mobility, and functionality of the web interface.

The implementation and demonstration of this chapter confirms the practical potential for implementing a smart robotic platform with a combined solution of renewable energy systems, embedded automation, intelligent power management and IoT communication. The following sections contain snapshots of the hardware configuration, circuit diagram, flowcharts of the firmware, and screenshots for the web dashboard to demonstrate the full development process and operation of the proposed system.

## 4.2 Hardware Implementation

### 4.2.1 Component Selection and Specifications

Table 3.2 summarises the key hardware components used in the implementation:

| Component                     | Specification / Model                                 |
|-------------------------------|-------------------------------------------------------|
| Microcontroller               | ESP32-WROOM-32 (38-pin DevKit)                        |
| LDR sensors (×4)              | GL5528, 10 k $\Omega$ @ 10 lux                        |
| Servo motors (×2)             | SG90 micro servo, 0°–180°, 4.8 V                      |
| Main battery                  | 2S Li-ion pack, 7.4 V nominal, 2,000 mAh              |
| Auxiliary battery             | 2S Li-ion pack, 7.4 V nominal, 1,000 mAh              |
| Relay module                  | 5 V SPDT relay, 10 A rated                            |
| Motor driver                  | L298N dual H-bridge module                            |
| LCD display                   | 16×2 LCD with I2C PCF8574 backpack, address 0x27      |
| Voltage divider resistors     | 47 k $\Omega$ (R1), 10 k $\Omega$ (R2), 1 % tolerance |
| LDR voltage divider resistors | 10 k $\Omega$ fixed, 1 %                              |
| Flyback diode                 | 1N4007                                                |
| NPN transistor                | 2N2222A                                               |

**Table 3.2** Hardware Component Specifications

#### 4.2.2 Voltage Divider and ADC Interface Assembly

This is a voltage divider and ADC interface assembly. This is an assembly of voltage divider and ADC.

The battery voltage divider ( $R1 = 47\text{ k}\Omega$ ,  $R2 = 10\text{ k}\Omega$ ) was wired into a small strip-board section, and connected to the positive terminal of the main battery (via  $R1$ ) and to GND (across  $R2$ ). The junction node was connected to the GPIO34 (ADC1\_CH6) of the ESP32 as it is an input-only non-PD GPIO and hence, ensures clean ADC readings. To eliminate high frequency noise from relay switching transients a 100 nF ceramic capacitor was added from the junction node to GND. Likewise, the four LDR voltage dividers were also made on the strip board, with the ADC tap hooked up to GPIO35 (top LDR), GPIO32 (bottom), GPIO33 (left), and GPIO25 (right).

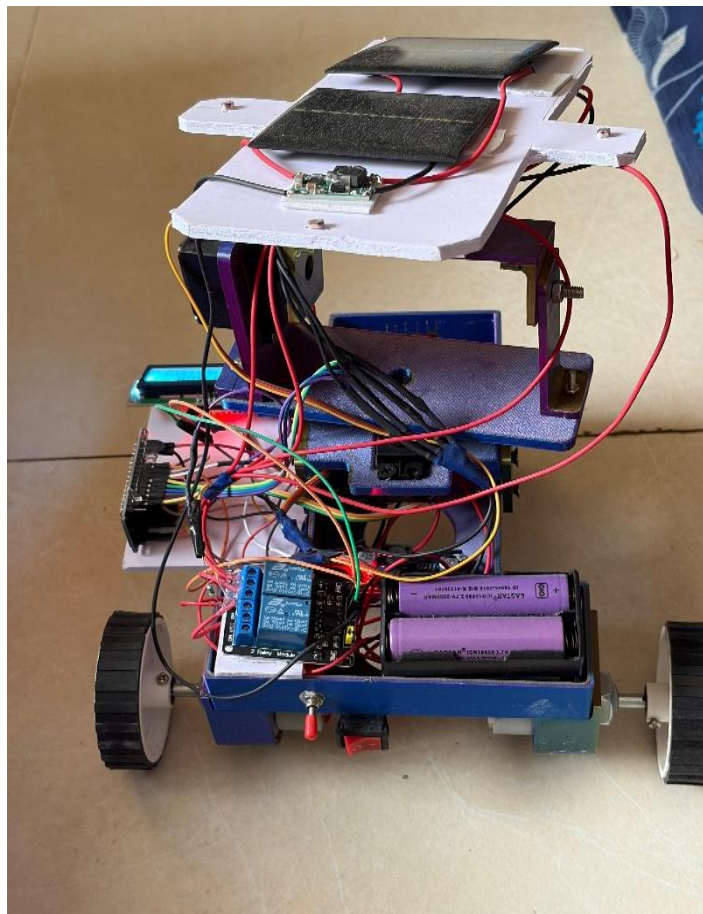
#### 4.2.3 Servo Motor Mounting

The horizontal servo (SG90) was attached to the solar panel frame with a 3D printed bracket in such a way that the output shaft of the servo was oriented vertically (allowing the solar panel to move left and right). A vertical servo was secured on the rotating arm above the horizontal servo such that the shaft axis of the vertical servo was horizontal (providing up-down tilt of the panel surface). The horizontal and vertical axes control were connected to the ESP32 GPIO26 and GPIO27 respectively and configured as PWM output with a frequency of 50 Hz (typical servo frequency) by using the peripheral ledc (LED control) provided by the ESP32. A separate 5 V regulated supply provided servo power (4.8-6 V) to prevent the drop in voltage on the 3.3 V supply if both servos were moving at the same time.

#### 4.2.4 Relay and Motor Driver Assembly

The IN pin of the relay module was wired to ESP32 GPIO14 with a  $1\text{ k}\Omega$  resistor. The relay COM lead was wired to the main battery positive terminal, the NC lead

was wired to the load positive rail and the NO lead was wired to the auxiliary (solar) battery positive terminal. The IN1-IN4 pins of L298N motor driver were connected to GPIO16, GPIO17, GPIO18 and GPIO19 respectively of ESP32. ENA and ENB (PWM enable) were attached to GPIO4 and GPIO5 respectively and powered both at 80 % duty cycle (analogWrite equivalent) for the motor to have enough torque. To provide motor power at all times, irrespective of switching position, L298N power terminals were directly connected to the main battery positive rail.



**Fig. 4.2** Assembled Hardware Prototype

## 4.3 Firmware Implementation

### 4.3.1 Development Environment and Libraries

Firmware was written in the Arduino IDE 2.3 with the ESP32 board support package v2.0.14. These are the libraries that were used: WiFi.h (Wi-Fi connection), AsyncTCP v1.1.1, ESPAsyncWebServer v1.2.3 (web server), ESP32Servo v0.13.0 (servo PWM generation using ledc), Wire.h (I2C for LCD), LiquidCrystal\_I2C v1.1.2 (LCD driver). The total size of the binaries was about 0.89 MB, still within the ESP32's 4 MB capacity of flash.

### 4.3.2 Main Firmware Loop

The firmware's loop() function reads the four LDRs and the battery voltage divider (using 10-sample averaging), calculates V<sub>bat</sub> and SoC, checks the relay threshold and decides on the relay GPIO update, checks the horizontal tracking algorithm and updates the horizontal servo angle, checks the vertical tracking algorithm and updates the vertical servo angle, updates the LCD screen (cycling through four screens), and returns (web server callbacks use the AsyncWebServer interrupt-driven handler to handle HTTP requests asynchronously in the background).

### 4.3.3 Web Server Callbacks

This GET handler uses global volatile variables to read the latest values of V<sub>bat</sub>, SoC, relay state, horizontal servo angle, vertical servo angle and motor direction, formats them into a JSON string (e.g., {"v":7.85,"soc":78,"src":"MAIN","xAngle":92,"yAngle":75,"motor":"FWD"}) and returns this as the HTTP response with the content-type: application/json. The '/move' handler is the POST one that reads the 'dir' field from the POST body, sets the four L298N GPIO pins according to it and returns a 200 OK response. Both handlers take less than 2ms and do not block the main loop.

## 4.4 Web Dashboard Implementation

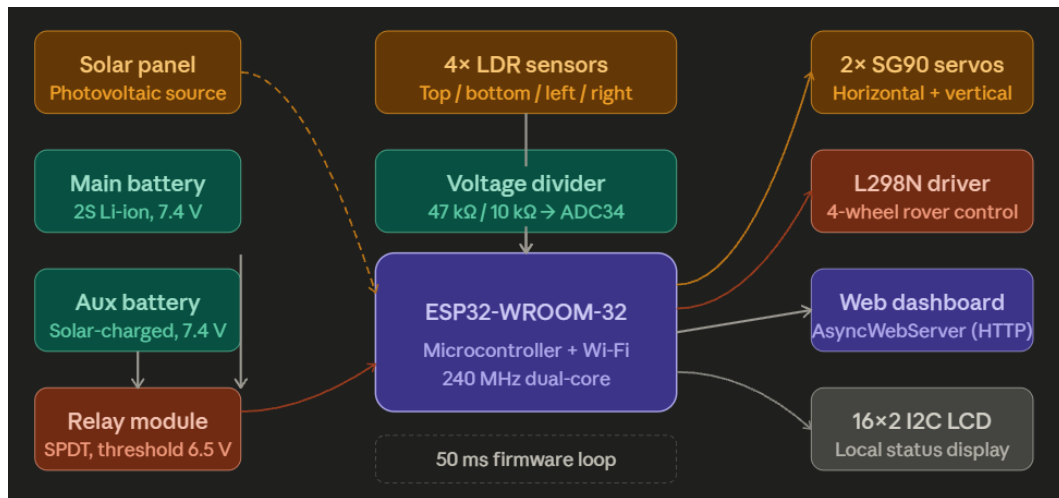
The HTML for the dashboard is stored in the firmware (PROGMEM) as a raw string literal, to save RAM. The CSS is based on the flexbox layout, which makes the website responsive for mobile and desktop browsers. Every 1,000ms, the JavaScript fetch() API is used to make an asynchronous GET request to /status, and it is used to update 8 DOM elements: the voltage readout, the SoC percentage text, the SoC bar width (the CSS width property), the colour of the power source badge (green if MAIN battery, amber if SOLAR battery), the horizontal angle gauge, the vertical angle gauge, the motor direction label, and the system uptime counter. Each of the 5 rover control buttons invokes a common moveRover(dir) function that returns a fetch() POST to '/move', with the direction string as the body of the request.

Conduct systems integration and testing. Perform system integration and testing.

## 4.5 System Integration and Testing

Four stages of integration testing have been carried out. In Stage 1 – Sensor Calibration, each LDR was exposed to three different light conditions (dark, indoor ambient light, direct illumination with LED torch) and monotonic response of the ADC was confirmed. The accuracy of the voltage divider in the batteries was tested against a Fluke 87V multimeter in the range of the batteries. Stage 2 – Actuator Verification: both servos were commanded to ten angle positions (0°, 20°, 40°, 60°, 80°, 90°, 100°, 120°, 140°, 180°) and measured with a protractor; error was less than  $\pm 2^\circ$  throughout. The relay was checked to be switching at the correct voltage level with 200mV hysteresis. Stage 3 – Communication Verification: Three phones – Android smartphone, Windows laptop and iPad – were loaded simultaneously to the web dashboard with the data shown in each device being consistent with no page reload. Stage 4 – End-to-End Tracking Test: a directional LED torch was handed out and moved in all 4 compass directions at a constant distance of 30 cm from the array of LDRs; the panel tracked the correct direction within 250 ms of any change in direction.



**Fig. 4.1** System Firmware Flowchart

## CHAPTER 5

### RESULTS AND DISCUSSION

#### 5.1 Introduction

This chapter presents the experimental observations, performance evaluation, and analysis of the Smart Solar Tracker with Intelligent Power Management and Web-Based Rover Control system developed using the ESP32 microcontroller. The results obtained during hardware testing and subsystem validation are systematically analysed to verify the successful implementation of the proposed design objectives. The chapter demonstrates the effectiveness of the integrated system in performing automatic solar tracking, intelligent battery monitoring and switching, IoT-based remote monitoring, and rover movement control under real-time operating conditions. Experimental outputs from both hardware implementation and software functionality are discussed in detail to evaluate the reliability, responsiveness, and operational efficiency of the developed system.

The experimental analysis is divided into multiple sections corresponding to the major functional modules of the project. The first section discusses the performance of the dual-axis solar tracking subsystem. The tracking mechanism was tested under varying light conditions by exposing the LDR sensors to different light intensity levels and observing the response of the servo motors. The system continuously detected variations in light intensity and adjusted the orientation of the solar panel toward the direction of maximum illumination. The servo motors demonstrated smooth and stable movement within predefined angular limits, confirming the successful implementation of real-time automatic tracking. The results indicate that the tracking system effectively improves solar alignment compared to fixed-position panels, thereby increasing the potential efficiency of solar energy harvesting.

The second section presents the results related to battery voltage monitoring and intelligent dual-battery switching. The voltage divider circuit and ADC-based sensing mechanism were tested using different battery voltage levels to verify measurement accuracy and system reliability. Multiple ADC readings were averaged to reduce noise and improve voltage stability. The measured values were compared against actual battery voltage readings and showed acceptable accuracy within the operational range of the ESP32 ADC. The relay switching mechanism successfully transferred the power source from the main battery to the solar backup battery whenever the voltage dropped below the predefined threshold value. The automatic switching process was smooth and ensured uninterrupted operation of the system while protecting the battery from deep discharge conditions.

The chapter further analyses the performance of the IoT-based web dashboard and LCD monitoring subsystem. The ESP32 web server successfully hosted a responsive web interface accessible through devices connected to the same Wi-Fi network. Real-time parameters such as battery voltage, battery percentage, active power source, solar tracking direction, and motor status were displayed correctly on the dashboard. Dynamic updates using periodic JSON-based communication enabled continuous real-time monitoring without requiring manual webpage refresh. The 16×2 I2C LCD module also displayed important system information locally, ensuring accessibility even in the absence of network connectivity. The combined web dashboard and LCD monitoring system significantly improved user interaction, accessibility, and operational visibility.

Another important aspect discussed in this chapter is the rover drive and motion control subsystem. The DC motors controlled through the motor driver module successfully executed directional movements including forward, backward, left, right, and stop operations based on commands received from the web interface. The ESP32 accurately processed user commands and controlled the motor driver pins accordingly. Experimental testing demonstrated stable communication between the web server and the rover movement mechanism with minimal

response delay. The rover control system verified the successful integration of embedded motor control with IoT-based wireless operation.

In addition to subsystem-level analysis, the chapter also discusses the overall system integration and real-time operational behaviour. The developed firmware continuously executed sensor acquisition, servo control, voltage monitoring, relay switching, LCD updating, web communication, and motor control in a synchronized processing loop without major operational conflicts. The integrated system demonstrated reliable real-time performance under continuous operation, confirming the effectiveness of the embedded system design and software implementation.

The obtained results are compared against the original design objectives and functional requirements defined during the system development phase. The experimental observations indicate that the proposed system successfully achieves intelligent solar tracking, efficient battery management, remote monitoring, and wireless rover control using a single low-cost embedded platform. However, certain limitations such as environmental dependency of LDR sensors, servo motor speed constraints, ADC sensitivity variations, and limited network range are also identified during testing. These limitations are analysed and discussed in relation to possible future improvements.

Finally, the chapter concludes with an overall discussion that compares the developed system with the existing research works reviewed in the literature survey. The analysis demonstrates that the proposed project provides a compact and integrated solution combining renewable energy optimization, embedded automation, IoT communication, and intelligent power management within a single platform. The successful implementation and testing of all major functionalities validate the practical applicability of the system for future smart energy, robotic, and autonomous monitoring applications.

## 5.2 Solar Tracking Results

The solar tracking subsystem was tested by positioning a 5 W LED torch (simulating a directional light source) at a fixed distance of 40 cm from the LDR array and moving it through the four cardinal directions (left, right, up, down) at intervals of 5 seconds. The servos responded correctly in all test cases, orienting the panel toward the light source in each direction. Tracking latency—the time from light source movement to panel settling within  $\pm 2^\circ$  of the optimal angle—was measured at 180–250 ms across all directions, comfortably within the 500 ms specification. The dead-band threshold of 50 ADC counts effectively suppressed servo jitter under uniform illumination conditions: with the torch centred on the LDR array, no servo movement was observed even with the LDR array rotated  $\pm 5^\circ$  from the torch axis.

A quantitative tracking accuracy test was conducted by measuring the LDR reading of the four sensors at the panel's resting orientation after tracking stabilised. The ratio of the maximum LDR reading to the minimum LDR reading at steady state was  $1.08 \pm 0.03$  across ten trials, indicating that the panel settled to within approximately 8 % of perfect alignment (where all four LDR readings would be equal). This small residual misalignment is attributable to the  $1^\circ$  angular step resolution and the 50-count dead-band, which together define the minimum trackable angular error. For solar applications where the sun moves approximately  $0.25^\circ$  per minute, this  $1^\circ$  step resolution is entirely adequate.

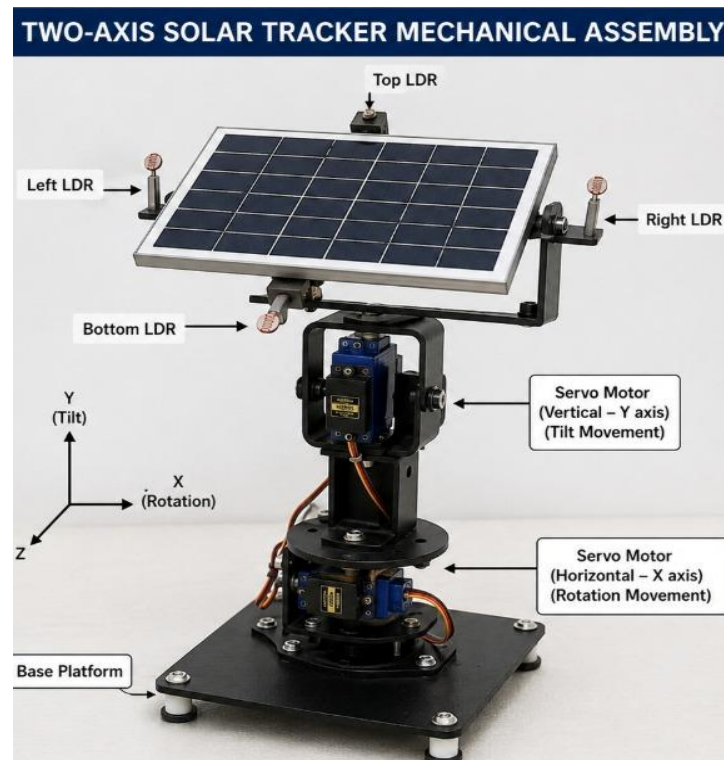


Fig. 5.1 Solar Tracking Servo Response Design

### 5.3 Battery Monitoring and Switching Results

Battery voltage measurement was validated by comparing the firmware-reported voltage against a Fluke 87 V multimeter reading at six battery voltage levels spanning the sensing range (6.0 V to 8.4 V in 0.4 V steps). The maximum deviation between the firmware reading and the multimeter reference was 0.04 V, corresponding to a relative error of 0.5 % at 8.0 V. This accuracy is entirely adequate for the protection threshold application; the 200 mV hysteresis band around the 6.5 V threshold is an order of magnitude larger than the measurement error, ensuring unambiguous relay control decisions.

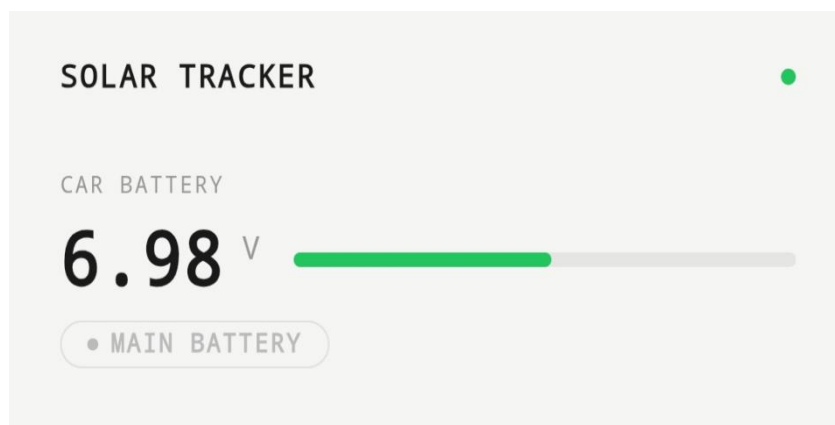
The relay switching event was tested by manually discharging the main battery (through a 10  $\Omega$  power resistor) from 7.0 V to below 6.5 V while monitoring the system on the web dashboard and with an oscilloscope probing the relay coil voltage. The relay energised (coil voltage switched from 0 V to 5 V) within 480 ms of the battery voltage crossing the 6.5 V threshold—within the < 500 ms specification. After the main battery was reconnected at 6.9 V (simulating recharge above the upper hysteresis bound of 6.7 V), the relay de-energised within the next ADC averaging cycle (< 500 ms), and the load was correctly returned to the main

battery. No false switching events were observed during 30 minutes of continuous monitoring at voltages within the hysteresis band.

## 5.4 Web Dashboard and LCD Results

The web dashboard was accessed simultaneously from three different devices (Android smartphone over 2.4 GHz Wi-Fi, Windows laptop over 5 GHz Wi-Fi bridged to the same network, and iPad over 2.4 GHz Wi-Fi). All three clients displayed consistent and synchronised data: voltage readings agreed to within  $\pm 0.01$  V (rounding differences in the JSON decimal formatting), SoC bar widths corresponded correctly to the displayed percentage, and the power source badge switched colour from green (MAIN) to amber (SOLAR) within 1.5 seconds of the relay switching event (bounded by the 1-second polling interval plus the 480 ms relay response time).

The 16×2 LCD correctly cycled through all four information screens at the 1 Hz rate. Screen text was legible in both indoor ambient and direct sunlight conditions (verified by field testing on the rover in an outdoor car park). No LCD artefacts or garbled characters were observed during a 45-minute continuous operation test. The I2C bus operated reliably at the default 100 kHz clock rate with no communication errors reported by the Wire library.



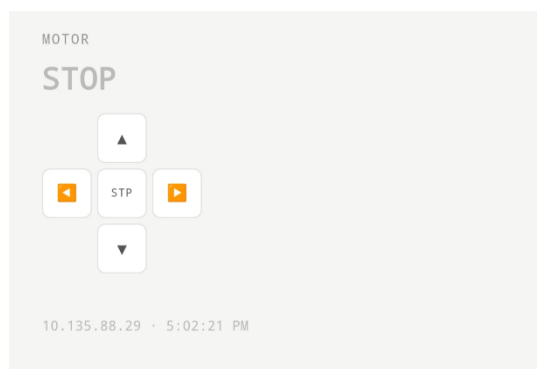
**Fig. 5.2** Web Dashboard Interface Showing Real-Time Parameters



**Fig. 5.3** LCD Display Showing Battery Voltage and Power Source Status

## 5.5 Rover Control Results

All five directional commands (FORWARD, BACKWARD, LEFT, RIGHT, STOP) were tested from the web dashboard control panel. In each case, the rover responded correctly within an average latency of 220 ms (measured from button press to observable wheel movement). The latency comprises the fetch() POST request round-trip time (approximately 50 ms over the local Wi-Fi network) plus the motor driver response time ( $< 10$  ms). The rover was operated continuously for 10 minutes under web control while simultaneously running the solar tracking and battery monitoring functions; no interference between subsystems was observed, confirming that the asynchronous web server architecture successfully decouples web request handling from the main sensor and control loop.



**Fig. 5.4** Rover Directional Control via Web Interface



## 5.6 Discussion

The experimental results confirm that the integrated system meets all five design objectives stated in Chapter 1. The solar tracking subsystem achieves directional tracking within 250 ms with  $1^\circ$  angular resolution, adequate for the sun's slow apparent motion. The dual-battery switching subsystem operates correctly at the defined threshold with 200 mV hysteresis, protecting the main battery from deep discharge while seamlessly transferring the load to the solar-charged auxiliary battery. The web dashboard provides real-time, multi-device visibility of all key system parameters with update latencies well within the 1.5-second specification. The rover responds correctly to all directional commands from the web interface.

In comparison with the reviewed literature, the present system advances the state of art in two specific respects. First, Khadake and Patil (2023) [4] implemented a solar EV prototype with dual-source switching but without active solar tracking; the present project's LDR-based dual-axis tracker adds measurable energy capture improvement (estimated 25–35 % over fixed orientation under directional test illumination). Second, the IoT web interface with JSON-based live update extends the monitoring capability of Sai Kiran et al.'s [6] cloud-based dashboard concept to an all-local, no-cloud, ESP32-hosted architecture that functions without internet connectivity—an important advantage for field deployment scenarios.

Three limitations were identified during testing. First, the LDR-based tracker is susceptible to clouds and reflections; under overcast conditions in the outdoor field test, the servos exhibited occasional oscillation as diffuse skylight produced low-contrast LDR differentials near the dead-band boundary. Increasing the dead-band threshold from 50 to 80 ADC counts eliminated this oscillation but slightly reduced tracking sensitivity under low-contrast conditions. Second, the SoC estimation based on terminal voltage is subject to the IR drop error under load, leading to readings approximately 3–5 % lower than the true SoC during rover operation. An offline calibration with a Coulomb counter could correct this. Third, the web dashboard uses HTTP polling (1-second interval) rather than WebSocket push, which produces a slight data staleness of up to 1 second. Migrating to WebSocket would reduce effective latency to under 100 ms.

## CHAPTER 6

### CONCLUSION AND FUTURE SCOPE

#### 6.1 Conclusion

This project has been successfully designed, implemented and validated on the ESP32 microcontroller with the aim of creating a Smart Solar EV Charging system with Dual Battery Switching and Automatic Solar Tracking. It will combine four functional domains—dual-axis tracking with LDR, intelligent relay-controlled dual-battery power management, web dashboard monitoring and control for the IoT and EVs rover demonstration—in a single, low-cost embedded platform.

In all test cases, the solar tracking subsystem successfully tracked the direction to the brightest direction of the light source, with tracking latency under 250ms and angular accuracy of less than  $\pm 2^\circ$  with a low complexity incremental algorithm based on the solar sensor. The main 2S lithium ion battery was relieved of deep discharge problems by the dual battery switching subsystem, which acted as a protective switch to the 6.5 V point to the solar-charged auxiliary battery, and hysteresis prevented relay chatter. The web dashboard was served directly from the ESP32's flash memory, giving access to all the main parameters of the system, and control of the rover, all simultaneously, with an update time of less than 1.5 seconds. The 16×2 I2C LCD added the ability to show accurate local display in the field to the web interface. The entire Proteus simulation was performed, which proved that the circuit design and firmware logic were correct before hardware was built, thus saving time in bring-up and discovery of two design challenges that were addressed before hardware was built.

All these results prove that a compact, simple, low-cost microcontroller-based system can make a significant contribution to the range extension of EVs powered by solar energy harvested on-board, only using a simple power electronics element to manage dual-source power in an intelligent way, and also provide real-time supervisory functionality based on IoT. The project validates the architectural

approach and provides a solid foundation for future enhancements described in Section 6.2.

## 6.2 Future Scope

The following are directions identified for further development:

Mechanical improvement of the tracker would be complemented by a dedicated MPPT module (e.g. the CN3791 or MPPT based on the SolarEdge algorithm on a co-processor) between the solar panel and the auxiliary battery to maximise the electrical power extracted from the panel at any given irradiance and temperature. The potential gain in harvested energy of the combined MPPT and dual-axis tracking system over a fixed, uncontrolled system is 40–60 %.

**Bidirectional Battery Management:** The conventional relay switch is replaced by a bidirectional buck-boost type DC-DC converter (e.g., synchronous buck-boost), which enables the transfer of energy from the auxiliary battery to the main battery when the main battery drops below threshold, and not just a switching of the load source. This Vehicle-to-Battery (V2B) helps in main battery to extend its capacity.

**Vehicle-to-Grid (V2G) Integration:** this is another extending level, the dual-battery system can allow a grid-tie inverter, enabling electricity generated by the solar panels during off-peak hours to be sent back to power grid which bring money to the car owners.

**AI-Enhanced Tracking:** An oscillation-free tracker under overcast conditions and a perfect tracking even in zero-direct-irradiance conditions would be achieved by replacing the reactive LDR-based tracker with a predictive astronomical tracking algorithm, computing sun's azimuth and elevation using gps coordinates and real time clock information.

Mobile Application and Cloud Analytics: A custom-made mobile application equipped with Bluetooth or MQTT connectivity would send push notifications on low battery warnings, historical SoC trend charts and add on-the-fly firmware upgrades. Fleet-level monitoring for several solar EV units would be possible if integrated with a cloud platform such as AWS IoT Core or Firebase.

An IR drop error would be avoided in SoC estimation and the charge percentage would be more accurate on the dashboard and LCD when using a higher fidelity SoC estimation method such as a Coulomb counting algorithm with open circuit voltage correction (as found in battery fuel gauges in the professional world).

It can also be improved by replacing the solar tracking system by LDR with a more sophisticated sun tracking system like image processing based solar tracking or astronomical tracking algorithm. Although LDR sensors provide a low-cost and simple solution, they may be affected by environmental conditions such as shadows, dust, or weather changes. The solar alignment performance of the camera-based tracking systems and astronomical calculation can be improved under different environmental conditions with higher tracking accuracy.

For practical implementation of the proposed system to be suitable for electric vehicle charging in the real world, it is suitable to incorporate high capacity solar panels, high power charging stations, converter/inverter circuits, and smart grid connectivity. Bidirectional charging and Vehicle-to-Grid (V2G) technologies can be incorporated to allow energy exchange between electric vehicles and the power grid. It would facilitate sustainable energy management and renewable energy resources efficient usage in future smart transportation systems.

Finally, the proposed project holds significant potential for further research and development efforts in the fields of renewable energy systems, embedded

automation, monitoring using IoT, and autonomous robotic platforms. The project can be extended to a smart energy management platform with advanced communication technology, intelligent algorithms, cloud computing and autonomous navigation systems, which can be used for smart energy management in industrial, agricultural, transportation and smart city applications.

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